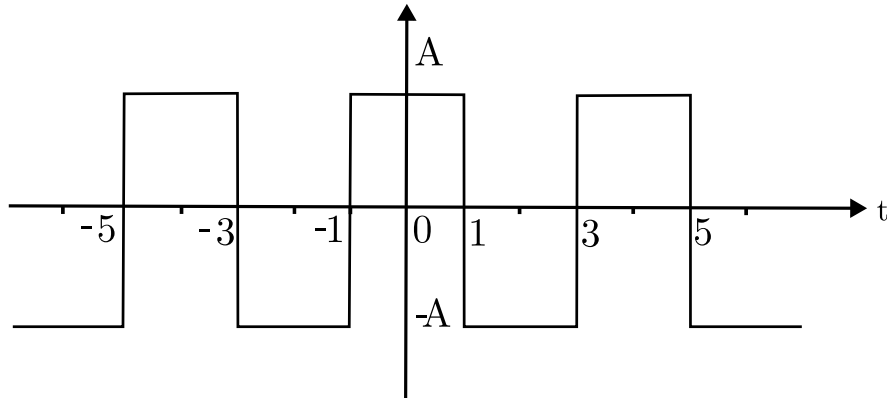


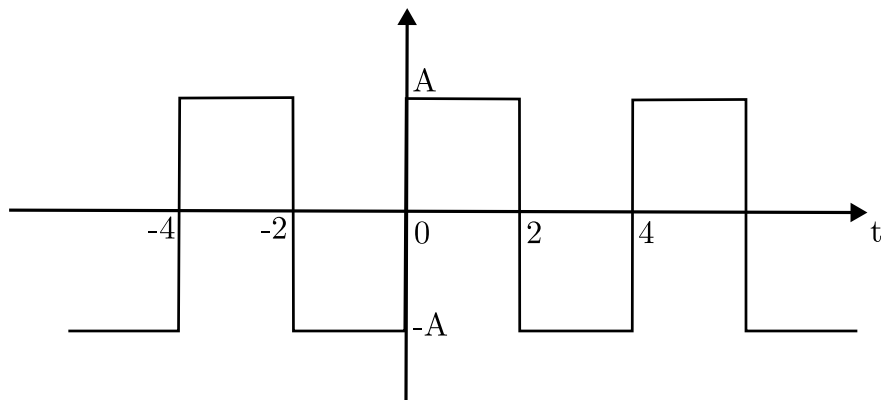
Tutorial 1: Fourier Analysis
EE 3203: Power Electronics

1. Identify symmetries (odd, even, half-wave), if any, in the following waveforms (without any changes to the axes):

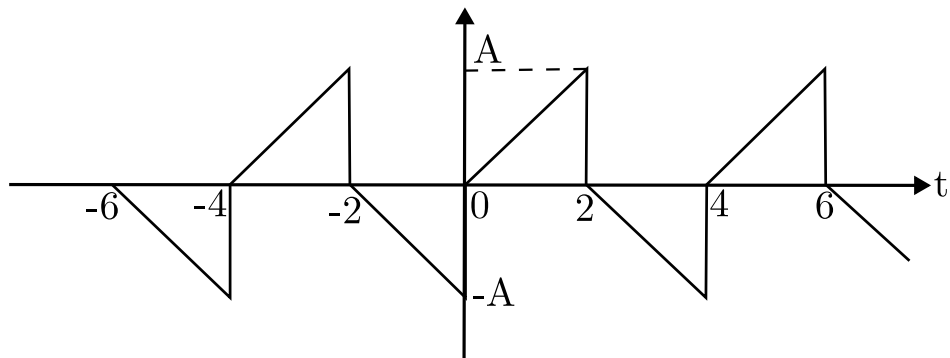
a.



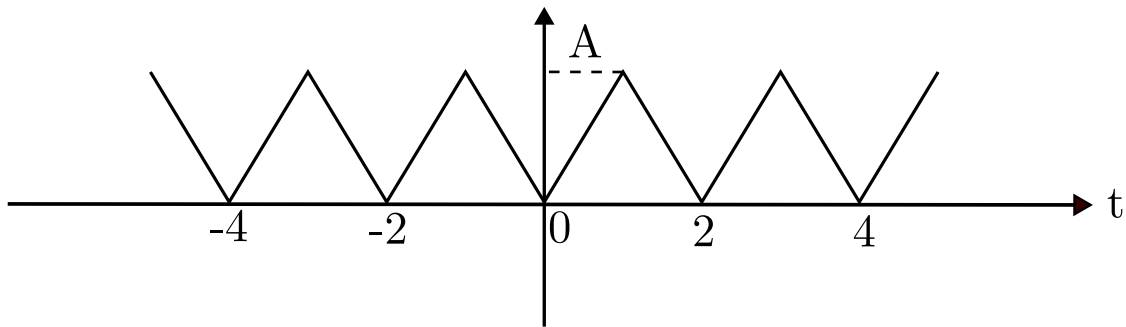
b.



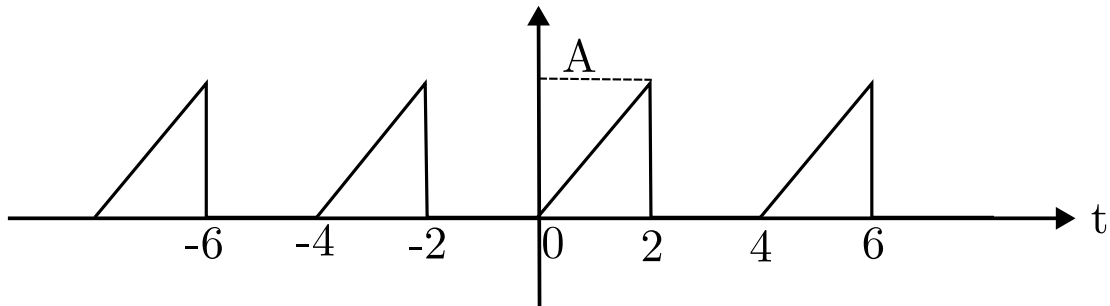
c.



d.



e.

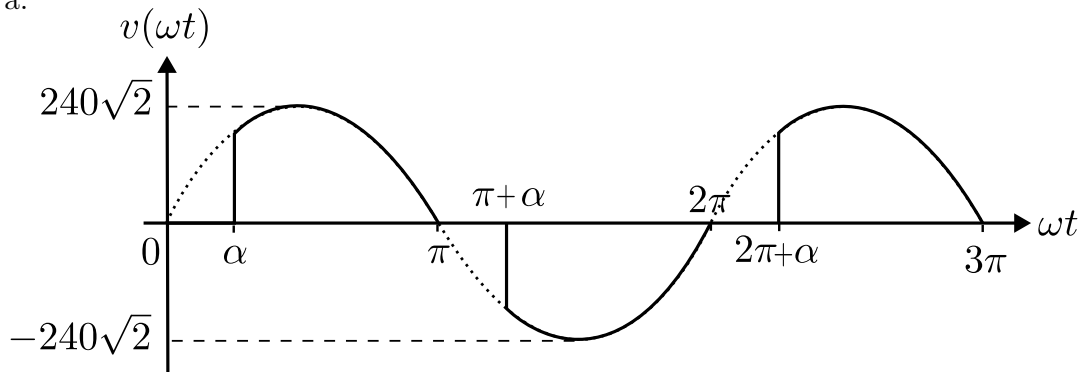


2. Find the Fourier series expansion of the following waveforms and express the same in the form

$$f(\omega t) = A_0 + \sum_{n=1}^{\infty} A_n \cdot \cos(n\omega t) + \sum_{n=1}^{\infty} B_n \cdot \sin(n\omega t)$$

Exploit the waveform symmetries wherever possible.

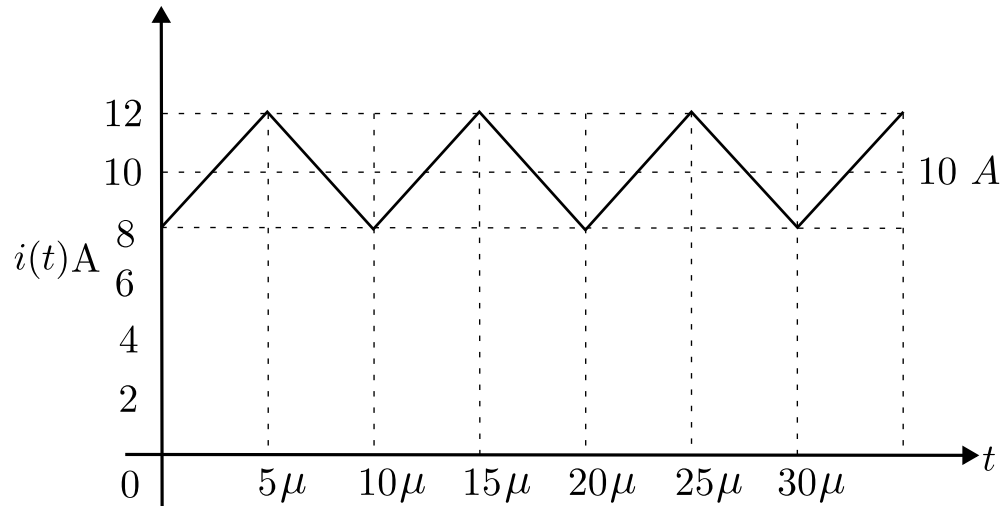
a.



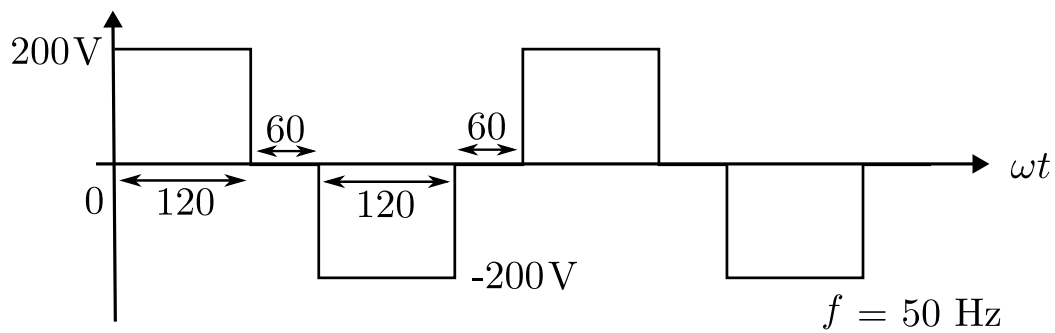
[The wave form is sinusoidal with some parts chopped as shown]

Tabulate the rms value of the fundamental frequency, 3rd harmonic and 5th harmonic components for $\alpha = 30^\circ, 45^\circ, 60^\circ$. Observe the variation of fundamental frequency component of voltage with angle α .

b.



c.



[Hint: Odd symmetry can be exploited by shifting the Y-axis]